

## AI Tools Used

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|----------------------|-----------------------------------------------------|
| <b>Date</b>          | 26 June 2026                                        |
| <b>Team ID</b>       | SWTID-2026-9014                                     |
| <b>Project Name</b>  | AdForge AI: AI-Powered Advertisement Copy Generator |
| <b>Maximum Marks</b> | 2 Marks                                             |

### AI Tools Used:

Documenting the specific artificial intelligence tools and platforms utilized during your project lifecycle helps evaluate efficiency and technological integration. Detail each AI tool, how it was applied, and the tangible benefit it provided to our workflow or final product.

| S<br>·<br>N<br>o | AI Tool / Platform Name                  | Specific Use Case / Application                                                                                                                                                                                                                                                                                                 | Project Phase Applied                                    | Impact / Benefit                                                                                                                                                                                                                                                        |
|------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                | <b><i>Groq API — Qwen3-32B Model</i></b> | <b><i>Core AI engine of AdForge AI. Used to generate all advertisement copy including Google Ads headlines and descriptions, Instagram captions with hashtags, Facebook primary text variations, marketing slogans, advertisement refinements, and performance intelligence reports. Integrated directly into the Flask</i></b> | <b><i>Development Phase &amp; Production Runtime</i></b> | <b><i>Reduced advertisement copy generation time from hours of manual writing to under 10 seconds per campaign. Enabled simultaneous generation of platform-specific content for three platforms in a single API call. Groq's LPU hardware delivered ultra-fast</i></b> |

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|   |                               | <i>backend via the Groq Python SDK.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         | <i>inference compared to standard GPU-based providers.</i>                                                                                                                                                                                                                                                    |
| 2 | <b>Claude AI (Anthropic)</b>  | <i>Used as the primary AI coding assistant throughout the entire project lifecycle. Generated the complete Flask backend including all routes, prompt engineering functions, input validation, safety filter, and Groq API integration. Also generated the full frontend including HTML templates, CSS stylesheet with premium SaaS design, and JavaScript controller. Additionally generated the automated test suite with 60 tests, Ngrok deployment script, README documentation, and all project planning documents.</i> | <b>Planning, Design, Development, Testing &amp; Documentation Phase</b> | Accelerated full-stack development by approximately 80%. Eliminated the need to write boilerplate code manually. Provided production-ready, secure, and well-commented code with proper error handling, MVC architecture, and modular structure. Reduced overall project development time from weeks to days. |
| 3 | <b>ChatGPT (OpenAI GPT-4)</b> | Used for research and ideation during the planning phase. Helped define the project scope, identify suitable marketing psychology frameworks such as AIDA, PAS, FAB, and BAB, research best practices for advertisement copywriting across Google, Instagram, and Facebook, and outline                                                                                                                                                                                                                                      | <b>Planning &amp; Research Phase</b>                                    | Provided comprehensive marketing domain knowledge that informed the prompt engineering strategy. Helped structure the project requirements clearly before development began, saving significant research time.                                                                                                |

|          |                       |                                                                                                                                                                                                                                                                                                                               |                                             |                                                                                                                                                                                                                                                 |
|----------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                       | <b>the feature list for AdForge AI.</b>                                                                                                                                                                                                                                                                                       |                                             |                                                                                                                                                                                                                                                 |
| <b>4</b> | <b>GitHub Copilot</b> | Used inside VS Code during local development and debugging to provide inline code suggestions, auto-complete Flask route handlers, assist with CSS styling rules, and suggest JavaScript event handler patterns. Also assisted in identifying and fixing the proxies TypeError caused by the groq and httpx version conflict. | <b>Development &amp; Debugging Phase</b>    | Sped up code writing inside VS Code by providing context-aware suggestions. Reduced debugging time for dependency conflicts. Helped maintain consistent code style throughout the project files.                                                |
| <b>5</b> | <b>Grammarly AI</b>   | Used to proofread and improve all written documentation including the README file, About page content, How It Works page content, project submission forms, demo script, team remarks, and all academic submission documents to ensure professional language, correct grammar, and formal tone.                               | <b>Documentation &amp; Submission Phase</b> | Ensured all submitted documentation met academic and professional writing standards. Eliminated grammatical errors and improved clarity of technical explanations. Maintained a consistently professional tone across all written deliverables. |